Hankel-Structured Channel Estimation for Rydberg Atomic Receivers: Bounds, Effective-Rank Scaling, and Out-of-Model Validation

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***Abstract*—Rydberg atomic receivers measure field magnitude only, making multi-user channel estimation a biased phase-retrieval problem in which unstructured solvers are flat in array size: their Fisher information is block diagonal across receive elements, so a larger aperture buys nothing under a normalised metric, and only a structural prior converts aperture into accuracy. We enforce the low-rank Hankel structure of the uniform-linear-array response inside an exact expectation–maximization Gerchberg–Saxton iteration, select the order from a held-out pilot residual, and benchmark against a Rician Cramér–Rao bound in which each magnitude measurement contributes rank-one information, together with a geometry-constrained bound on the $3\sum_k L_k$-parameter manifold. The gain grows from -0.19 to $+2.85$ dB as N goes $8 \to 32$ while the unstructured baseline moves by 0.012 dB. We then show the governing variable is not path count but effective rank relative to capacity, and use that relation to predict the gain on two channel models it was never fitted to. Simulation only; the bounds do not govern biased estimators.**

***Index Terms*—Rydberg atomic receiver, channel estimation, phase retrieval, Hankel matrix, constrained Cramér–Rao bound, effective rank.**

I. Introduction

Atomic receivers based on Rydberg states read out radio fields through an optical transition whose response is proportional to field *magnitude*, discarding phase [1], [2]. Multi-user channel estimation is therefore biased phase retrieval, and Gerchberg–Saxton-type alternating projection [3] is the natural solver; recent work unrolls it into a trained network [4], and a rank projection has been placed inside projected gradient descent for the same problem [5].

Unstructured solvers share a limitation that is easy to miss and, once seen, explains a great deal: **their accuracy does not improve as the array grows.** Row n of $\mathbf{GS}$ enters only measurement row n, so the Fisher information is block diagonal across receive elements and the per-element estimation problem is unchanged by adding elements. Aperture is only useful to an estimator that couples elements — which is exactly what a structural prior does.

On a uniform linear array (ULA) each propagation path contributes one spatial complex exponential, so the Hankel lifting of every channel column is rank deficient [6], [7]. Enforcing that inside the iteration turns aperture into accuracy. But knowing *when* it applies has, until now, meant a qualitative appeal to sparsity, and that is unusable: a 16-path channel on a 32-element array is algebraically full rank yet has effective rank 8.73, and its lifting is compressible with nothing sparse about it.

Contributions.

1) A benchmark, not just a baseline: a Rician Cramér–Rao bound in which each magnitude measurement contributes *rank-one* information, and a geometry-constrained bound in tangent-space form because the parameterisation is rank deficient. EM-GS sits within 0.05 dB of the unconstrained bound, so the benchmark is credible.
2) The aperture-scaling result and its Fisher-information explanation.
3) Replacement of the path-count heuristic by $r_{\mathrm{eff}}/r_{\max}$, with the capacity $r_{\max} = \lceil N/2 \rceil$ derived.
4) Two out-of-model predictions, each registered before measurement, on channel models the relation was never fitted to — the clustered-propagation falsification test that this work's earlier stage identified as decisive.

II. System Model

An N-element ULA serves K single-antenna users over P pilots. With $\mathbf{G} \in \mathbb{C}^{N\times K}$, pilots $\mathbf{S}$, known reference beam $\mathbf{B}$ and noise $\mathbf{W}$, the receiver observes

$$\mathbf{Z} = |\mathbf{GS} + \mathbf{B} + \mathbf{W}|, \qquad \mathbf{W}_{np} \sim \mathcal{CN}(0, \sigma^2), \tag{1}$$

elementwise, and the modulus is never linearised. User k's channel is

$$\mathbf{g}_k = \sum_{\ell=1}^{L_k} \alpha_{k\ell}\, \mathbf{a}(\theta_{k\ell}), \qquad [\mathbf{a}(\theta)]_n = e^{-j\pi n \sin\theta}, \tag{2}$$

so each column is a sum of complex exponentials in the element index.

RSR convention, stated once. The reference-to-signal ratio is defined against a *single* user. Since K users transmit at once, a nominal single-user RSR of 12 dB is $12 - 10\log_{10} K = 7.23$ dB against the K-user total at $K = 3$; 10 dB single-user is 5.23 dB. **[TODO: An earlier draft of this work stated 7.15 dB for the 12 dB case; the arithmetic gives 7.23 dB and the 0.08 dB difference could not be reconciled from any stored result. Confirm which is intended.]** All RSR values quoted here are single-user.

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Two averaging domains, kept distinct. Within an operating point NMSE is pooled as a ratio of sums,

$$\mathrm{NMSE} = 10\log_{10}\frac{\sum_t \|\hat{\mathbf{G}}_t - \mathbf{G}_t\|_F^2}{\sum_t \|\mathbf{G}_t\|_F^2}, \tag{3}$$

never a mean of per-trial decibels: (3) estimates the ensemble quantity a Cramér–Rao bound bounds, whereas a mean-of-decibels is smaller by Jensen. Across SNR, the mean is of per-SNR decibel values, after conversion; linear NMSE spans three orders of magnitude, so a linear average across SNR is dominated by the lowest-SNR point and nearly blind to the high-SNR half. **Rule used throughout:** ratio-of-sums wherever a curve is compared to a bound; paired per-trial median with a bootstrap interval for paired contrasts. Both are reported; neither is used for the other's job.

III. Hankel Structure and the HS-GS Estimator

For $\mathbf{g} \in \mathbb{C}^N$ and pencil p, let $\mathcal{H}_p(\mathbf{g})$ be the $(N-p)\times(p+1)$ Hankel matrix. Under (2) its rank is $\min(L_k, N-p, p+1)$, so the largest expressible rank is

$$r_{\max}(N) = \max_p \min(N-p,\, p+1) = \lceil N/2 \rceil, \tag{4}$$

attained at $p = \lfloor N/2 \rfloor$. The ceiling, not the floor: at $N = 7$ the maximum is 4, and the two expressions coincide only for even N. A constraint at $r = r_{\max}$ is satisfied by every vector and carries no information, which fixes the scale any rank must be read against.

K **cancels.** Unstructured, $\mathbf{G}$ has $2NK$ real degrees of freedom; under (2) it has $3\sum_k L_k$. With $L_k \equiv \bar{L}$,

$$\frac{3\sum_k L_k}{2NK} = \frac{3K\bar{L}}{2NK} = \frac{3\bar{L}}{2N}, \tag{5}$$

independent of K. Since the projection acts per user column, this predicts K-invariance of the structural gain at fixed pilot adequacy $P/2K$.

The estimator. HS-GS interleaves, after every exact EM-GS update, a Cadzow projection $\mathcal{P}_r(\mathbf{g}) = \mathcal{H}_p^\dagger(\Pi_r(\mathcal{H}_p(\mathbf{g})))$, where Π_r truncates to r leading singular values and $\mathcal{H}_p^\dagger$ averages antidiagonals. One application is *not* idempotent: it is one step of an alternating projection between the rank set and the Hankel set, not a projection onto their intersection, so the number of sweeps is part of the operator's definition and is stated with every result below. The exact step is the EM-GS update throughout, so HS-GS and HS-EM-GS name the same estimator; we use HS-GS to match the released code.

Order selection uses no oracle. At each operating point a subset of pilots is held out, the estimator is run at reduced iteration count for every candidate $r \in \{1, \ldots, r_{\max}\}$, and the r minimising the held-out residual is selected. The rule also *abstains*: when it selects $r \ge r_{\max}$ the projection is the identity and HS-GS reduces exactly to EM-GS.

The governing variable. We index results by the Roy–Vetterli effective rank [8] **[TODO: VERIFY CITATION]** of the noiseless column's lifting, $r_{\text{eff}} = \exp(-\sum_i p_i \ln p_i)$ with $p_i = \sigma_i / \sum_j \sigma_j$, taken relative to $r_{\max}$. Unlike L, this is a property of any channel rather than of one generator, and it is what makes the out-of-model predictions of §V-D possible. It must be computed on the noiseless channel: the effective rank of the *estimate* is set by the noise floor and collapses every configuration onto one value.

IV. Performance Benchmark

A. Rank-one Fisher information

Conditional on $\mathbf{G}, \mathbf{S}, \mathbf{B}$ the noiseless field at (n,p) is a constant λ, so $z = |\lambda + w|$ is Rice distributed and its density depends on λ *only through* $a = |\lambda|$. With $\frac{d}{dx}\log I_0(x) = R(x)$ and $\kappa = 2za/\sigma^2$, the score is $\partial_a \log p = \frac{2}{\sigma^2}[zR(\kappa) - a]$; writing $\beta = \sigma^{-4}(\mathbb{E}[z^2R^2(\kappa)] - a^2)$, one measurement carries scalar information 4β, so in real coordinates $\boldsymbol{\eta} = [\Re\mathrm{vec}\,\mathbf{G}; \Im\mathrm{vec}\,\mathbf{G}]$,

$$\mathbf{J}_n = \sum_{p=1}^{P} 4\beta(a_{n,p}, \sigma^2)\, \mathbf{u}_{n,p}\mathbf{u}_{n,p}^T. \tag{6}$$

Each magnitude measurement contributes rank-one information, not rank two. Crediting both quadratures overstates it and gives a bound too low by 0.17–4.64 dB across our 36 points. An identifiability check settles the form: at $K = 3$, $P = 3$ the per-row problem is unidentifiable and (6) is correctly singular, whereas the rank-two construction claims full rank.

Because row n of $\mathbf{GS}$ enters only measurement row n, $\mathbf{J}$ is *block diagonal across receive elements*. This is the reason an unstructured estimator, and any estimator attaining the unconstrained bound, is flat in N under a normalised metric — the result §V-B measures.

B. Constrained bound on the geometric manifold

The unconstrained bound governs estimators treating $\mathbf{G}$ as $2NK$ free parameters. HS-GS exploits (2), whose parameters $\boldsymbol{\phi} \in \mathbb{R}^{3\sum_k L_k}$ number about 45, *independent of* N. The textbook form $\mathbf{D}(\mathbf{D}^T\mathbf{J}\mathbf{D})^{-1}\mathbf{D}^T$ with $\mathbf{D} = \partial\boldsymbol{\eta}/\partial\boldsymbol{\phi}$ is unusable, $\mathbf{D}$ being rank deficient: at $N = 8$ its mean numerical rank is 40.4 against ≈ 44.7 parameters, and 28.1% of realisations have $3\sum_k L_k > 2NK$, so $\boldsymbol{\phi} \mapsto \mathbf{G}$ is not injective. We use the Gorman–Hero / Stoica–Ng tangent-space form [9], [10] **[TODO: VERIFY CITATION (both)]** with $\mathbf{U}$ an orthonormal basis of $\mathrm{range}(\mathbf{D})$:

$$\mathrm{CCRB} = \mathbf{U}(\mathbf{U}^T\mathbf{J}\mathbf{U})^{-1}\mathbf{U}^T, \tag{7}$$

which depends on the tangent *space* and is invariant to over-parameterisation. Both bounds average over the same realisations as the estimator curves.

Bias caveat. Both bounds constrain the covariance of *unbiased* estimators. All estimators here run to a fixed budget with a shrinking update, and HS-GS additionally truncates rank, so none is unbiased and neither bound applies strictly: the unconstrained CRLB is a reference for GS and EM-GS only, and the CCRB does not forbid a sufficiently biased structured estimator from falling below it.

TABLE I
CONFIGURATIONS. EVERY FIGURE IS INTERNALLY SINGLE-CONFIGURATION; NONE MIXES POINTS FROM THE TWO.

	Bound comparisons	Effective-rank study
N	8, 16, 32	16, 32, 64
K / P	3 / 30	3 / 20
L_k	$\mathcal{U}\{3, 7\}$ [5]	$\mathcal{U}\{3, 7\}$ [5]
SNR	-5 to 20 dB	$\mathcal{U}[-10, 20]$ dB
RSR (single-user)	12 dB	10 dB
Iterations T	50	100
Cadzow sweeps	4	1
Statistic	ratio-of-sums	paired median, CI
Trials	3.6×10^4 total	83–874 per cell

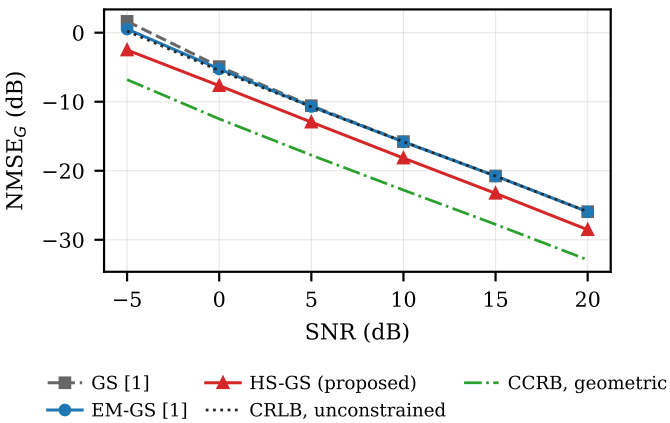

Fig. 1. NMSE versus SNR, bound-comparison configuration ($N = 32$, $P = 30$, RSR 12 dB, $T = 50$, 4 Cadzow sweeps, ratio-of-sums). EM-GS tracks the rank-one unconstrained CRLB to within 0.05 dB above 5 dB. Neither bound governs a biased estimator.

V. NUMERICAL RESULTS

Table I lists both configurations. The older is canonical for everything compared to a bound, because it is what the bounds were computed against; the effective-rank and out-of-model results are reported at their own configuration, and each figure names the one it uses.

A. Main result against the bounds

At $N = 32$, $P = 30$, RSR 12 dB (Fig. 1), EM-GS tracks the unconstrained CRLB to within 0.05 dB for SNR ≥ 5 dB, so it is close to efficient for the unstructured problem and the benchmark is credible; at -5 and 0 dB the curves separate by 0.30 and 0.24 dB, consistent with low-SNR bias. The geometric CCRB lies 0.87–9.98 dB below the unconstrained bound, quantifying what the structure is worth in principle, and HS-GS occupies part of that gap, improving on EM-GS by 2.27–3.04 dB.

B. Aperture scaling

Fig. 2 isolates the scaling. Averaged over twelve operating points per array size the gain is -0.19, $+0.78$, $+2.85$ dB at $N = 8, 16, 32$, with win rates 33.5, 74.0, 95.3% and the constraint active in 57.5, 92.7, 99.6% of trials. **Under the same averaging EM-GS varies by** 0.012 **dB**, exactly as the block-diagonal Fisher structure of §IV requires: the N-dependence belongs to the structural prior, not to a degrading baseline.

$N = 8$ is a mixed result, not a small positive one. There $r_{\max} = 4$ while $L_k \in \{3, \ldots, 7\}$, so the projection is inactive in 43% of trials and where active often truncates real components; HS-GS gains $+0.744$ dB at -10 dB SNR but loses 0.190–0.403 dB above 0 dB, and the sign of any scalar summary depends on the pooling. Truncation bias does not shrink with noise, so as variance falls the trade stops paying.

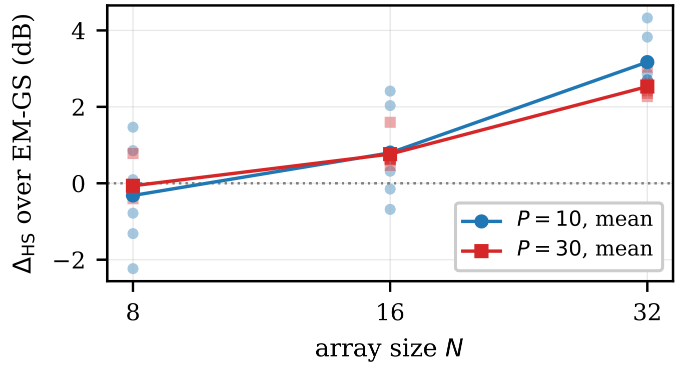

Fig. 2. Paired HS-GS gain over EM-GS versus array size, bound-comparison configuration. Faint markers are the six individual SNR operating points, solid lines their mean. At $N = 8$ the points straddle zero, so no single scalar summarises that array size.

C. From path count to effective rank

Fixing $L_k = L$ across users and sweeping to the ceiling, the gain decays strictly monotonically, 7.04, 3.56, 1.79, 1.04, 0.58, 0.27, 0.05, -0.12 dB for $L = 2, 4, \ldots, 16$, with EM-GS flat to 0.19 dB across the sweep — which rules out generic denoising, since that would persist at large L where the estimate is equally noisy rather than vanish exactly where the prior becomes vacuous.

Re-indexing that sweep by $r_{\text{eff}}/r_{\max}$ generalises it. The zero crossing moves from $L/r_{\max} = 0.90$ to $r_{\text{eff}}/r_{\max} = 0.518$, and the useful region becomes $r_{\text{eff}}/r_{\max} \lesssim 0.5$ — a statement about any channel rather than about one simulator's L values. Fig. 3 places $N \in \{16, 32, 64\}$ on that axis. The crossing is 0.588, 0.518, 0.544 across a fourfold change in aperture, a spread of 0.070; the $N = 16$ and $N = 64$ values are extrapolated from the last two measured points rather than bracketed.

The gain magnitude does not collapse onto the same axis, and we state the residual rather than report a collapse. Over the window both arrays cover, $N = 16$ lies above $N = 64$ by a mean of 0.493 dB and a maximum of 0.901 dB, one-signed throughout and monotone in N. We have no account of it. One candidate is pencil quantisation at small capacity — $r_{\max}(16) = 8$ admits eight ranks against 32 at $N = 64$ — but that is untested.

D. Out-of-model prediction: the falsification test

An earlier stage of this work concluded that robustness under clustered propagation was not established and named

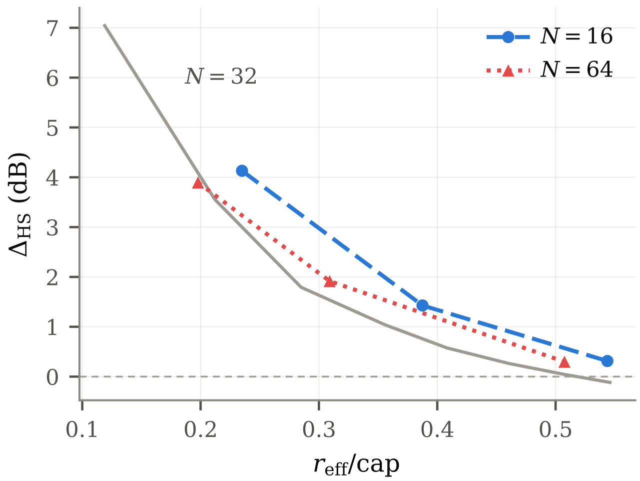

Fig. 3. Δ_{HS} against $r_{\mathrm{eff}}/r_{\max}$, effective-rank configuration (Table I, right column; adaptive order, 1 Cadzow sweep, paired median). The boundary is invariant to within 0.070 across $N \in \{16, 32, 64\}$; the magnitude is not, with a one-signed residual of mean 0.493 dB (max. 0.901 dB).

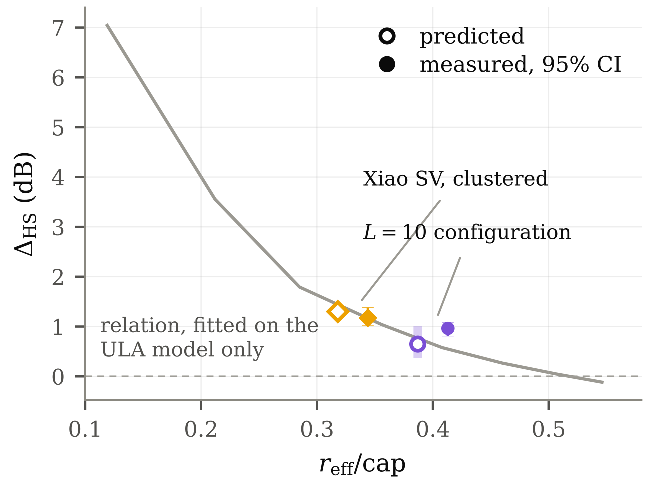

Fig. 4. Predictions registered before measurement, against measurement, effective-rank configuration. The relation is fitted on the geometric ULA model of (2) only; neither channel was used to fit or tune it.

the decisive next experiment: *repeat the principal comparison under a clustered generator — a falsification test, not a confirmation exercise.* This is that test, run twice. In each case $r_{\mathrm{eff}}/r_{\max}$ was computed, a predicted Δ_{HS} read off the relation, and **the prediction committed to version control before the measurement was run**; the relation was not refitted afterwards.

Clustered Saleh–Valenzuela. At $r_{\mathrm{eff}}/r_{\max} = 0.331$ the relation predicts $+1.30$ dB. Measured: $+1.173$ dB, CI $[+1.017, +1.380]$ — an error of 0.13 dB, prediction inside the interval.

A second, independently specified configuration. We adopt the receive-side Saleh–Valenzuela configuration of [11] — $L = 10$ paths, $\alpha_\ell \sim \mathcal{CN}(0,1)$, angles $\mathcal{U}(-90°, 90°)$. **[TODO: FILL IN AUTHORS of arXiv:2408.14366 from page 1 of the PDF; it is not readable from the environment these results were produced in]** That work is a *precoding* study, not a competing estimator, and only its channel configuration is borrowed; it uses ULAs at both ends with incident and departure angles, of which we use the receive side only. At $r_{\mathrm{eff}}/r_{\max} = 0.400$ the relation predicts $+0.648$ dB, interval $[+0.367, +1.015]$. Measured: $+0.963$ dB, CI $[+0.808, +1.086]$. Inside the registered interval and on the correct side of zero, but the point estimate is 0.315 dB low — a larger error than the first, and we report it as such rather than as a second 0.13 dB hit. Together the two bound what the relation is good for: it places an unseen channel on the correct part of the curve, and it is not accurate to a tenth of a decibel in general.

Clustering did not eliminate the gain, which was the outcome the earlier stage identified as possible. On a third reading in which all 40 rays are drawn independently, $r_{\mathrm{eff}}/r_{\max}$ rises to 0.748, the relation predicts -0.12 dB, and the measured median is exactly 0.000 dB. The magnitude is right to 0.12 dB but *the mechanism is not the predicted one*: rather than degrading, the order-selection rule declines to project at all in 29.1% of trials against 1.4% on the clustered model, without being told which channel it faces. We report that as a partial match.

E. Users, and a replication

The K-invariance predicted by (5) holds at SNR ≥ 5 dB, where Δ_{HS} across $K \in \{2, 3, 4\}$ at fixed $P/2K$ spans 0.095 dB. Pooled over all SNR it does not: the spread is 0.294 dB and monotone in K, missing a pre-registered falsifier of 0.3 dB by 0.006 dB, which we do not present as a pass. The pooled trend is driven by bins below 0 dB where each cell contributes about 45 trials.

The two configurations are a replication. At $N = 32$ the same contrast measures $+2.85$ dB under the bound-comparison configuration and $+2.680$ dB under the effective-rank one — different code, different RSR, iteration count, Cadzow sweep count and pooling rule, agreeing to 0.17 dB.

VI. Limitations and Conclusion

A sparse geometric ULA channel makes each column's Hankel lifting rank deficient, and this can be enforced inside an exact biased phase-retrieval iteration without linearising the magnitude observation or estimating any angle. Because the Fisher information is block diagonal across receive elements, the unstructured baselines are flat in array size and HS-GS alone converts a larger aperture into lower normalised error: $-0.19 \to +2.85$ dB as N goes $8 \to 32$ against 0.012 dB of baseline movement. The governing variable is effective rank relative to capacity, not path count, and that relation predicted the gain on two channel models it was never fitted to — one to within 0.13 dB, the other inside a registered interval but 0.32 dB off. Clustered propagation, identified earlier as the decisive falsification test, did not eliminate the gain.

Limitations. All results are simulation only, with no measured hardware and no recorded channel data. They cover the geometric ULA model of (2) and the Saleh–Valenzuela configurations of §V-D, and no claim is made beyond those families; the gain magnitude does not collapse onto the proposed axis and its residual is unexplained; the benefit is conditional and negative at $N = 8$ above 0 dB. A single scalar order is imposed on all K users despite their differing path counts, the projection step is approximate on a non-convex set with no convergence guarantee, and equal user power leaves near–far effects untested. Finally, both bounds constrain unbiased estimators, and every estimator here is biased.

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